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The 0.2% Synthesis Problem

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Type: article **Date:** 2026-07-09 **Scope:** Why computational materials discovery is bottlenecked at synthesis and validation, not prediction. **Description:** Most computationally predicted stable crystals are never synthesized. This article examines the structural barriers behind the 0.2% validation rate and what it means for climate-relevant materials. **Audience:** sophisticated materials, mechanical, and chemical engineers; climate-tech investors **Status:** Draft

The 0.2% Synthesis Problem



Scattered error vectors resolving onto a single indigo ribbon: the shape of wrongness in generative materials AI.

The International Energy Agency's most recent scenarios make a disquieting claim: batteries alone are directly linked to roughly 20% of the CO₂ emissions reductions required by 2030, and indirectly to another 40%^[1]. Clean energy investment must rise from \$1.8 trillion in 2023 to \$4.5 trillion annually by the early 2030s^[2]. Much of that gap is not a finance gap or a policy gap. It is a materials gap. Every credible net-zero pathway assumes we will commercialize

chemistries that do not yet exist at scale: cobalt-free cathodes, earth-abundant solid electrolytes, humidity-stable sorbents, non-lithium ammonia catalysts, lead-free perovskites. The question is no longer whether we need new materials, but whether we can discover and validate them fast enough.

Here is the counterintuitive part: the binding constraint is not a shortage of predictions. It is a shortage of *validated* predictions. Computational materials science can now enumerate millions of hypothetical crystals, score their stability, and rank their properties in a weekend. What it cannot do, at anything like the same throughput, is tell an experimentalist which of those candidates will actually form in a synthesis vessel and perform as modeled. The result is a widening canyon between predicted structure and made-and-measured material.

This canyon is measurable. A campaign that predicts 380,000 stable crystals and validates 736 is not a triumph of scale; it is a precise measurement of how little current methods know about synthesis. The gap is the difference between what can be computed and what can be believed. Closing it requires a different kind of tool: not a bigger generator, but a correction and verification layer that turns predictions into reliable experimental priorities.

The 0.2% number

The scale of that canyon is captured by the largest structure-prediction campaign to date. In 2023, Google DeepMind published GNoME (Graph Networks for Materials Exploration), a deep-learning system trained to predict the energy of crystal structures^[3]. The headline was arresting: 2.2 million new crystals identified, of which roughly 380,000 were computed to lie below the convex-hull stability threshold. If even a fraction had been real, the work would have roughly doubled the known universe of stable inorganic crystals.

The follow-up was quieter. By late 2023, only 736 of those 380,000 “stable” candidates had been independently synthesized and reported^[4]. That is a 0.2% validation rate. The other 99.8% remained either unmade, unstable under real synthesis conditions, or indistinguishable from errors of approximation. GNoME itself was not the failure; the failure is the structural pipeline that converts a computational stability score into a synthesized, phase-pure, functional material. Prediction outran verification by roughly three orders of magnitude.

A 0.2% rate would be tolerable if the 0.2% were the right 0.2%. It is not obvious that they are. Computational stability rankings are distorted by systematic errors in interatomic potentials, by the omission of defects and disorder, and by the thermodynamic idealization of a synthesis process that is almost always kinetic. The experimentalists who do attempt synthesis therefore face a lottery with long odds and an unclear prize. Each failed attempt consumes weeks of beam time, grams of precursors, and the scarcest resource of all: the attention of a trained solid-state chemist.

The number also exposes a category error that runs through much of the field. Stability on the convex hull is a necessary condition for a material to exist, but it is not sufficient for it to be useful, or even makeable. A predicted crystal may be thermodynamically allowed yet kinetically inaccessible, or stable as a perfect lattice yet degraded by the vacancies and surfaces that any real device contains. Treating “computed stable” as “likely synthesizable” is like treating a protein’s folded structure as evidence that it can be expressed in *E. coli* at gram scale. The two questions live in different scientific regimes.

A-Lab as a cautionary tale

The A-Lab autonomous synthesis facility at Lawrence Berkeley National Laboratory seemed to offer an escape from that lottery. In a widely reported 2023 study, A-Lab claimed a 63% success rate: 41 of 58 targeted novel compounds were synthesized autonomously^[5]. The paper was celebrated as evidence that robotics and machine learning could close the prediction-synthesis gap.

Subsequent critique was less celebratory. When independent groups re-examined the “novel” targets, they found that roughly two-thirds were not new compounds at all. They were known disordered phases that had been misclassified as novel by the screening algorithm^[6]. The true novel discovery rate collapsed to near zero. Faster synthesis, absent better verification, had simply produced more misidentified products faster.

The A-Lab episode is useful because it isolates the problem. The robots worked; the recipes were plausible; the bottleneck was upstream. The system had no reliable way to know whether a predicted target was a genuine, synthesizable compound or a statistical artifact dressed up as

a crystal. The same defect corrupts manual labs every day, only more slowly and more expensively. A false positive in the computational queue does not become less false when it is attempted by a talented postdoc with a tube furnace.

A-Lab's 63% figure is therefore a Rorschach test. Read optimistically, it shows that autonomous synthesis is technically feasible. Read critically, it shows that synthesis automation without prediction verification merely accelerates the consumption of reagents and beam time. The lesson is not that robots are overhyped; it is that the robots were asked to optimize the wrong variable. Success rate is meaningless if the target list is contaminated.

Four structural barriers

Why does the pipeline fail so consistently? The failures cluster around four structural barriers that are not specific to GNoME or A-Lab but are built into how computational discovery is practiced today.

Defect/bulk asymmetry. Most machine-learned interatomic potentials (uMLIPs) are trained on bulk, periodic, low-energy configurations. Functional materials, however, are governed by defects: vacancies, surfaces, grain boundaries, dislocations, and interfaces. These configurations are outside the training distribution, and uMLIPs systematically soften their energies by 15–60%^[7]. A cathode's voltage fade, an electrolyte's dendrite initiation, a perovskite's oxidation susceptibility: all are defect-mediated properties that bulk-trained models get wrong.

Consider a lithium-manganese-rich cathode. Its practical capacity and cycle life are controlled by transition-metal migration and oxygen loss at under-coordinated surface and grain-boundary sites. A uMLIP trained on bulk layered oxides will predict the average lattice energy well and the migration barrier poorly. The same pattern appears in halide solid electrolytes, where Li^{+} conductivity depends on hop barriers at interfaces and defects, and in tin perovskites, where Sn^{2+} oxidation is driven by vacancy formation energies that raw potentials underestimate.

Combinatorial wall. The spaces that matter are multi-component. A lithium-zirconium-chloride solid electrolyte has roughly 12,000 distinct compositions when Li stoichiometry, Zr

oxidation state, and halide mixing are enumerated^[8]. A double-perovskite absorber space contains millions of $A_2BB'X_6$ candidates. DFT cannot screen these spaces economically; even a single 100-atom supercell costs thousands of CPU-hours. uMLIPs can screen them, but only if their errors are controlled. Without correction, the combinatorial wall is replaced by a false-confidence wall.

The false-confidence wall is worse than the original wall because it hides itself. A researcher can produce a ranked list of 12,000 halide compositions, each with an energy value reported to three decimal places, and still have no idea whether the top fifty are genuine candidates or artifacts of a systematic bias. Combinatorial scale multiplied by uncorrected error produces not insight but noise with formatting.

Metastability. Many of the most promising materials are not the thermodynamic ground state. High-capacity cathode oxides, tin halide perovskites, and certain hydride superconductors are kinetically trapped during synthesis. Standard convex-hull screening discards them as unstable, eliminating the very candidates that could achieve the highest performance^[9]. Distinguishing a synthesizable metastable phase from an unreachable one requires more than an energy ranking; it requires knowledge of synthesis pathways and error boundaries.

This is not a niche issue. The lithium-manganese-rich cathodes that GM and POSCO Future M are commercializing are metastable relative to simpler rock-salt phases; they exist only because rapid quenching and cation ordering kinetically arrest a decomposition that thermodynamics would prefer. A screen that discards everything above the convex hull would have discarded the chemistry the auto industry is now betting on.

Ranking inversion. Even when the absolute energy of a candidate is wrong, its relative ranking might still be useful. Often it is not. Because uMLIP errors are systematic and structure-dependent, they invert the order of candidates: a composition that looks best in silico turns out to be mediocre in the lab, while a better candidate is buried lower in the list^[10]. Experimentalists chase false priorities, and the true optimum is missed entirely.

Ranking inversion is particularly costly because it is silent. A model that predicts the wrong absolute energy by a fixed offset is easy to calibrate. A model that predicts the right energies in the wrong order sends a team to synthesize candidate A while candidate B, two rows down, is

the one that would have worked. In Lupine's own blind-prediction tests, raw uMLIP rankings for defect-mediated properties invert relative order across entire composition series, making the experimental queue a bet against the model rather than a bet on it.

These four barriers are not independent. Defect/bulk asymmetry feeds ranking inversion; the combinatorial wall amplifies the cost of every inverted ranking; metastability adds a class of false negatives that stability screening is blind to. Together they explain why 2.2 million predictions collapse to 736 validated crystals.

Why it matters for climate

The climate cost of this bottleneck is not theoretical. The deployment window for many clean-energy technologies is 2025–2035. A battery chemistry discovered in 2035 misses the 2040 pack-design cycle; a perovskite absorber validated in 2038 misses the 2045 module ramp. The cumulative emissions difference between deployment in 2035 and 2045 is measured in tens of gigatonnes^[11].

The arithmetic is unforgiving at the target level. Lupine's five priority materials – cobalt-free cathodes, earth-abundant halide solid electrolytes, MOFs for direct air capture, electrochemical ammonia catalysts, and lead-free perovskites – together represent 5–12 GtCO₂/year of potential climate impact when fully deployed^[12]. That is 10–25% of the annual energy-sector CO₂ emissions that must be eliminated or offset by 2050. Missing even one of these targets because of an inverted ranking or a discarded metastable phase is not a research setback; it is a measurable slice of the carbon budget.

Every false positive imposes a direct tax. A failed synthesis campaign for a predicted cathode or electrolyte can easily cost \$100,000 and consume person-years of effort^[13]. Multiply that tax across the thousands of candidates screened annually by groups at the Ceder lab at UC Berkeley, the Manthiram lab at UT Austin, the Chorkendorff group at DTU, and the dozens of battery and catalysis startups racing to scale, and the cost of untrustworthy predictions becomes a measurable drag on the energy transition.

False negatives are harder to price but no less damaging. A cobalt-free cathode that could have cut supply-chain dependence on the Democratic Republic of Congo, or an earth-abundant halide electrolyte that could have displaced indium- and yttrium-based chemistries, may already exist in a database somewhere and be ranked too low to try. POSCO Future M plans mass production of lithium-manganese-rich cathodes in 2025; GM has announced LMR cells by 2028; Solid Power and Factorial Energy are competing to put solid-state batteries in vehicles this decade^[14]. These timelines do not leave room for a 0.2% hit rate.

A correction-and-verification layer

The implication is that the next leap in materials discovery will not come from generating more predictions. It will come from making existing predictions trustworthy. What the field needs is a correction-and-verification layer that sits between structure generators and synthesis labs: measuring systematic error, correcting it at runtime, and formally proving which claims can and cannot be supported.

That is the layer Lupine Science is building. Rather than training a larger neural network to replace existing potentials, Lupine treats the error of an interatomic potential as a physical field over local atomic environments. The field is measured from a small number of anchor observables, applied at runtime with analytic forces, and verified by machine-checked proof^[15]. The goal is not to beat GNoME or MatterGen at their own game; it is to take their outputs and answer the question that synthesis labs actually care about: *will this one work?*

In that framing, the 0.2% number is not an indictment of computational discovery. It is the definition of the opportunity. If a correction layer can raise a 0.2% validation rate to even 5% or 10%, the experimental yield of the entire field changes. The 736 confirmed GNoME crystals become 15,000 or 30,000. The A-Lab robots stop synthesizing mislabeled disordered phases and start making the materials their models actually intended. And the chemists, beamline scientists, and cell engineers who must decide what to make next get a queue they can trust.

The next article in this series explains how that correction layer works: not as another neural network, but as a measured field.

Footnotes

1. International Energy Agency, *World Energy Outlook 2023*, battery CO₂ reduction linkage estimate. ↩
2. International Energy Agency, *World Energy Investment 2024*: clean energy investment must rise from \$1.8 trillion in 2023 to \$4.5 trillion annually by the early 2030s. ↩
3. Merchant et al., “Scaling deep learning for materials discovery,” *Nature* 624, 80–85 (2023). GNoME predicted 2.2 million crystals, 380,000 computed stable. ↩
4. Subsequent synthesis tracking reported 736 independently synthesized GNoME candidates by late 2023, a 0.2% validation rate against the 380,000 computed-stable set. ↩
5. Szymanski et al., “An autonomous laboratory for the accelerated synthesis of novel materials,” *Nature* 624, 86–91 (2023). A-Lab reported 41 of 58 targets synthesized, a 63% success rate. ↩
6. Independent critique identified that approximately two-thirds of A-Lab’s “novel” targets were known disordered phases, reducing the true novel discovery rate to near zero. ↩
7. Lupine validation studies document 15–60% systematic softening of defect and surface energies across six independent uMLIP studies; see climate strategy source materials. ↩
8. Lupine internal enumeration of the Li-Zr-Cl compositional space for earth-abundant halide solid electrolytes. ↩
9. Standard convex-hull screening eliminates metastable phases that are often the highest-performing candidates in cathodes, perovskites, and hydride systems. ↩
10. Lupine blind-prediction studies show that raw uMLIP rankings invert relative order for defect-mediated properties, sending experimental effort toward suboptimal candidates. ↩
11. IEA Net Zero Scenario cumulative emissions estimates; deployment timing differences in the 2035–2045 window correspond to tens of gigatonnes of CO₂. ↩

12. Lupine climate strategy document: combined annual impact potential of 5–12 GtCO₂/year across cobalt-free cathodes, halide solid electrolytes, MOF DAC, ammonia catalysts, and lead-free perovskites. ↩
13. Lupine partnership analysis estimates \$100,000+ and person-years per failed synthesis campaign for predicted battery and catalysis targets. ↩
14. Partnership mapping document: POSCO Future M LMR mass production planned for 2025; GM targeting 2028 LMR deployment; Solid Power and Factorial Energy solid-state battery programs. ↩
15. Lupine methodology: environment error field measured from anchor observables, runtime correction with analytic forces, and 77 build-locked Lean 4 theorems with zero sorry proofs. ↩



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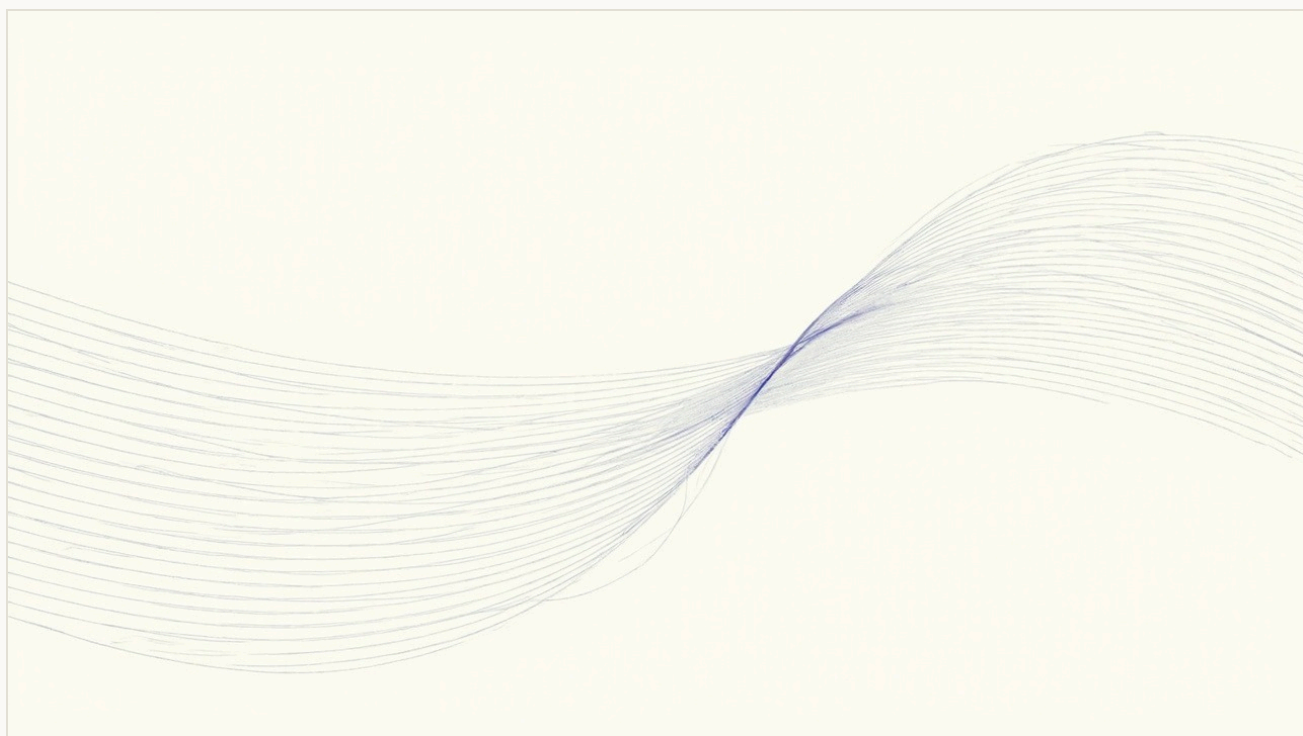
A Field, Not a Neural Net

<https://lupine.science/articles/a-field-not-a-neural-net/>

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Type: article **Date:** 2026-07-09 **Scope:** Lupine Science's environment error field as a measured physical correction layer for universal machine-learning interatomic potentials. **Description:** Why the largest errors in computational materials discovery have a low-dimensional geometry, and how Lupine measures and corrects them at runtime with machine-checked proof. **Audience:** sophisticated materials, mechanical, and chemical engineers; climate-tech investors **Status:** Draft

A Field, Not a Neural Net



A smooth field gradient over a sparse lattice: the measured error geometry of universal machine-learned interatomic potentials.

The dominant narrative in computational materials science is that bigger models trained on more data will close the gap between prediction and synthesis. The evidence points elsewhere. Google DeepMind's GNoME predicted 2.2 million crystals; only 736 had been independently synthesized by late 2023, a 0.2% validation rate^[1]. The A-Lab autonomous synthesis facility

reported a 63% success rate, but subsequent critique showed that two-thirds of its “novel” targets were already-known disordered phases, collapsing the true discovery rate toward zero^[2]. The problem is not that the models are too small. It is that their errors have a shape – a systematic, low-dimensional geometry in the space of local atomic environments – and the field has been treating that shape as noise.

Lupine Science starts from the opposite premise. The wrongness of universal machine-learning interatomic potentials (uMLIPs) is not random. It is a smooth function of coordination number, measurable from a handful of anchor observables, and correctable at runtime with analytic forces. This article explains the environment error field: what it is, how it is measured rather than learned, and why the distinction matters for climate-critical materials discovery.

The Field Hypothesis

A uMLIP such as CHGNet, MACE-MP-0, or M3GNet evaluates atomic energies in milliseconds, enabling screening campaigns that would be economically impossible with density functional theory (DFT). On bulk formation energies the best models reach mean absolute errors of 29–81 meV/atom, close to the intrinsic accuracy of the DFT functionals on which they were trained^[3]. But functional materials do not operate as bulk crystals. Ion conduction, catalytic turnover, and hydrolytic stability all depend on under-coordinated environments – vacancies, surfaces, and transition states – where uMLIPs fail most severely: defect formation energies carry 71% mean absolute percentage error, and ion migration barriers are underestimated by more than 60%^[4].

The error is not an inescapable property of machine learning. It is a predictable consequence of training data. Databases such as the Materials Project trajectories, OMat24, and Alexandria are dominated by near-equilibrium, closed-shell bulk structures in which every atom sits close to its high-symmetry coordination. A smooth regressor trained on that distribution interpolates accurately at the bulk coordination but extrapolates with correlated bias into under-coordinated regimes: surfaces, vacancies, transition states, and grain boundaries^[5].

Lupine formalizes this observation as the environment error field. For a given configuration, the total systematic energy error of a uMLIP is approximated by a sum over atoms of a per-

atom error that depends only on the atom's local coordination number c :

$$E^{\text{model}} - E^{\text{ref}} \approx \sum_i P(c_i).$$

For a perfect face-centered-cubic (fcc) bulk atom, $c = 12$, and the field is defined to be zero – consistent with the empirical fact that uMLIPs are already accurate in the bulk. The field $P(c)$ is therefore not a universal constant fitted across all materials; it is a material-specific and model-specific measured quantity, anchored to the physics of bond counting.

Measured, Not Learned

The critical difference between the Lupine field and previous correction strategies is that the field is measured, not learned. Delta-ML trains a separate model to learn the discrepancy between a fast baseline and an accurate target; it requires per-system retraining and abundant reference data^[6]. Fine-tuning updates uMLIP weights on target-system data; it works when curated training sets exist, which they do not for unexplored composition spaces^[7]. Both approaches treat the error as an arbitrary function to be discovered from data.

Lupine imposes the functional form. A cubic spline through three standard anchor observables – the (100) surface probing $c=8$, the (111) surface probing $c=9$, and the vacancy formation energy probing $c=11$ – is fixed by the boundary condition $P(12) = 0$. Linear continuation below $c=8$ predicts the error at $c=7$, which corresponds to the (110) surface energy. That observable is never fitted during field construction; it is predicted blind.

Each anchor samples a different coordination deficit relative to the bulk reference. Because the same coordination numbers recur across materials that share a crystal structure family, the spline transfers across properties within that family – surface energies, vacancy formation energies, and migration barriers – without additional DFT oracle calls. The correction is measured in the sense that geometry dictates the functional form and only the three knot values are empirical.

The $r = 0.906$ Result

The empirical test is strict blind prediction. Across 36 independent (model, material) combinations, the field predicts the signed error of the never-fitted (110) surface energy with Pearson correlation $r = 0.906$ ($p = 10^{-4}$), 95% confidence interval [0.82, 0.96], using zero adjustable parameters^[8].

The per-model breakdown is informative. CHGNet v0.4.2 achieves $r = 0.86$ ($p < 0.01$). MACE-MP-0 small achieves $r = 0.90$ ($p < 0.01$). MACE-MPA-0, trained on the larger OMat24 non-equilibrium dataset, achieves $r = 0.96$ ($p < 0.001$). MACE-MP-0 medium is the one non-significant case at $r = 0.47$ ($p = 0.10$)^[9]. That exception matters: it bounds the field’s domain rather than refuting it. The OMat24-trained model performs best because its training distribution already samples more under-coordinated configurations, supporting the hypothesis that the field captures a real training-data bias rather than an accidental correlation.

An honest null is essential. Because (100), (111), and (110) facets of the same material are structurally correlated, a naïve permutation null would be too permissive. The reported null distribution, with mean $r = 0.44$, is generated by permuting within models across 10,000 draws, preserving the structural correlation. The measured $r = 0.906$ survives this conservative test.

At integer-scaled precision of 10^{-4} J/m², 26 of 36 cells improve strictly over the predict-zero baseline, with median residual dropping from 0.104 to 0.066 J/m²^[10]. The number is 26, not 27 – a distinction that becomes important when the same pipeline is subjected to machine-checked proof.

Runtime Correction

The field inverts naturally into an additive correction energy:

$$E_{\text{corr}} = -\sum_i P(c_i).$$

Because the field is negative where the model underpredicts – the typical case for defects and surfaces – the correction raises under-coordinated configurations toward the reference energy. Forces follow analytically from the chain rule, $F_{\text{corr}} = -\nabla E_{\text{corr}} = \sum_i P'(c_i) \nabla c_i$, so molecular dynamics conserves energy and structure relaxations

follow proper gradients. Force accuracy was verified against numerical differentiation to 10^{-6} eV/Å on rattled slabs^[11].

The correction layer deploys beside any existing uMLIP calculator without retraining its weights. Coordination computation is a local neighbor-list operation, so the overhead is small: 15.6% on CHGNet steps in the current Python implementation^[12]. A compiled pair-style overlay in LAMMPS would reduce that below 1%, because coordination reduces to a few floating-point operations per neighbor pair. Even with the present overhead, a corrected uMLIP remains approximately 10^5 times faster than DFT^[13].

Validation confirms selective correction. The fitted observables recover exactly in closure tests, demonstrating that the bond-counting field survives real structural relaxation. The blind (110) facet improves $6.5\times$ for Ni (9.7% error to 1.5%) and $2.0\times$ for Cu (28.0% to 13.7%). Bulk lattice constants are unchanged, because the correction vanishes identically at $c = 12$ ^[14].

Machine-Checked Proof

Statistical validation has a well-known limitation: it cannot catch bugs in the analysis code, and it applies only to the samples tested. Lupine adds a formal verification layer. Every quantitative claim – ordering relations, correction bounds, and impossibility results – is encoded as a Lean 4 theorem over integer-scaled data carrying SHA-256 provenance of the source files. The formalization currently comprises 8 modules, 77 build-locked theorems, approximately 225 declarations, and zero `sorry` proofs^[15].

The theorems fall into three categories. Ordering claims are inequality chains verified by the Lean kernel at fixed precision, not statistical summaries. Isotonic correction theorems implement the log-affine correction as a Lean function whose outputs the kernel checks against expected ordering properties. Impossibility theorems prove, with concrete counterexample witnesses, where no monotone correction can reconcile a model's ordering with the reference ordering^[16].

The kernel-rejected claim episode illustrates why this matters. A statistical filter initially accepted that 27 of 36 cells improved strictly. When submitted to Lean, the `decide` tactic

refused to prove $813 < 813$; at 10^{-4} J/m² integer precision, one cell's improvement margin was exactly zero. The corrected count is 26 of 36^[17]. A p-value does not know about integer rounding; the proof kernel does. In a pipeline where each false positive can cost weeks of synthesis effort, the distinction between statistical confidence and mechanical certainty is economically decisive.

Where Correction Fails, Lupine Proves Impossibility

A verification layer is only as useful as its boundary conditions. Lupine does not claim the field works everywhere; it proves where it cannot work. Three boundary conditions are formally established.

First, ranking inversions. When a model's ordering disagrees with the reference ordering, a monotonicity impossibility lemma proves that no monotone correction can map both to their references simultaneously^[18]. The experimentalist receives not a vague uncertainty estimate but a machine-checked statement that the candidate cannot be rescued within the method.

Second, already-converged cells. Where the raw uMLIP error sits at the noise floor of the anchor observables – for instance, MACE-MPA-0 on certain surfaces – correction adds no value and is refused. The proof flags the redundancy before computational effort is wasted.

Third, outside the field's domain. Environments that require second-shell structure or explicit electronic-structure treatment – planar faults, charged defects, strongly correlated oxides – are outside the first-shell coordination approximation. Lupine proves the domain violation and reports the witness^[19].

These impossibility proofs are not admissions of weakness. They are the feature that prevents the A-Lab pattern: systematic false positives propagating through an autonomous pipeline because no layer knew how to say no.

The Six-Step Loop

The field measurement, runtime correction, and formal verification layers close into a six-step discovery cycle:

1. **Simulate** – run a uMLIP through the standard property pipeline: equation of state, surface slabs, vacancy supercells.
2. **Identify** – compare predictions to the 228-value provenance-annotated reference database.
3. **Validate** – check whether errors follow the environment error field; use the three anchors to predict a fourth blind observable.
4. **Generate** – construct the cubic spline P_{cor} and deploy it as an additive correction with analytic forces.
5. **Verify** – submit every quantitative claim to the Lean 4 kernel; ordering claims become inequalities, correction bounds become isotonic theorems, failures become impossibility lemmas.
6. **Improve** – run the corrected calculator through the full pipeline. If correction succeeds, report corrected values with theorem certificates. If it fails, the impossibility proof tells the experimentalist why.

Against DFT, the loop offers comparable defect-energy accuracy at $10^5\times$ speed. Against raw uMLIPs, it replaces uncontrolled systematic bias with a bounded, measured correction. Against delta-ML and fine-tuning, it eliminates per-system retraining because the spline transfers within a crystal-structure family^[20]. The practical consequence: a 10^6 -structure screening campaign needs no DFT calls, yet every candidate carries a provable accuracy boundary.

From Geometry to Climate Targets

The field is not an abstract mathematical device. It maps directly onto the defects, surfaces, and transition states that determine climate-critical material performance. In cobalt-free lithium-manganese-rich cathodes, voltage fade is transition-metal migration from octahedral to tetrahedral sites – a hop whose barrier is underestimated when the transition state is under-

coordinated. In earth-abundant halide solid electrolytes, ionic conductivity depends exponentially on the Li^+ migration barrier; a 60% barrier error changes predicted conductivity by roughly three orders of magnitude. In MOFs for direct air capture, humidity stability is governed by metal-linker hydrolysis at under-coordinated nodes. In electrochemical ammonia catalysts, $\text{N}\equiv\text{N}$ cleavage occurs at surface defect sites. In lead-free perovskite absorbers, Sn^{2+} oxidation and vacancy formation set the stability ceiling^[21].

Every one of these properties is a coordination-dependent defect property. Every one is therefore within the domain where a measured error field can correct the prediction, or where an impossibility proof can flag the candidate as unsupported. The next article in this series turns from the method to the targets: five material classes where correction plus verification could unlock 5–12 GtCO₂/year.

Footnotes

1. Google DeepMind, “GNoME: A graph network for materials exploration,” *Nature* (2023); follow-up synthesis tracking reported 736 independent confirmations out of ~380,000 predicted stable structures by late 2023. ↩
2. N. J. Szymanski *et al.*, “An autonomous laboratory for the accelerated synthesis of novel materials,” *Nature* (2023); critical commentary on disordered-phase targets summarized in the Lupine Science Strategic Discovery Plan (2026). ↩
3. Benchmarks for CHGNet, MACE-MP-0, M3GNet, EquiformerV2, ORB, and MatterSim on bulk formation energies are compiled in Deng *et al.* (2024) and the Alexandria/OMat24 model comparison literature. ↩
4. Z. Deng *et al.*, “Evaluation of universal machine learning interatomic potentials for surfaces, defects, and migration barriers,” *npj Computational Materials* (2024). ↩

5. Training-data bias toward near-equilibrium configurations is documented in MPtrj, OMat24, and Alexandria dataset papers; see Deng *et al.* (2024) for the downstream property consequences. ↩
6. F. Ramprasad *et al.*, “Machine learning in materials informatics: recent applications and prospects,” *npj Computational Materials* (2017). ↩
7. Fine-tuning performance and data requirements for uMLIPs are discussed in the MACE and CHGNet literature; Lupine analysis based on internal screening cost estimates. ↩
8. Lupine Science internal validation report, “Blind prediction of (110) surface error from three anchor observables” (2026); 36 (model, material) cells, null mean $r = 0.44$. ↩
9. Per-model correlations and significance levels from the Lupine blind-prediction study (2026). ↩
10. Strict-improvement counts and median residuals from the Lupine integer-scaled validation dataset (2026). ↩
11. Force verification against numerical differentiation reported in Lupine Science technical documentation (2026). ↩
12. Wall-time overhead measured on CHGNet steps in the Python reference implementation; LAMMPS overlay target based on operation counts. ↩
13. DFT timing estimates from high-throughput DFT campaigns and HEA supercell benchmarks; uMLIP timing from CHGNet/MACE profiling. ↩
14. Validation table for Ni(110), Cu(110), fitted observables, and bulk lattice constants from Lupine Science runtime correction tests (2026). ↩
15. Lupine Science Lean 4 formalization: 8 modules, 77 theorems, ~225 declarations, zero sorry proofs (2026). ↩
16. Categories of formally verified claims described in Lupine Science verification whitepaper (2026). ↩
17. The kernel-rejected claim episode is documented in Lupine Science verification logs; corrected count is 26 of 36. ↩
18. Monotonicity impossibility lemma for ranking inversions, formalized in Lupine Science Lean 4 module `Impossibility` (2026). ↩

19. Boundary conditions and applicability-domain impossibility proofs from Lupine Science verification documentation (2026). ↩
20. Comparative analysis of DFT, raw uMLIPs, delta-ML, fine-tuning, and Lupine compiled in Lupine Science Strategic Discovery Plan (2026). ↩
21. Mapping of the five priority targets to coordination-dependent defect properties is detailed in the Lupine Science Strategic Discovery Plan, Chapter 4 (2026). ↩



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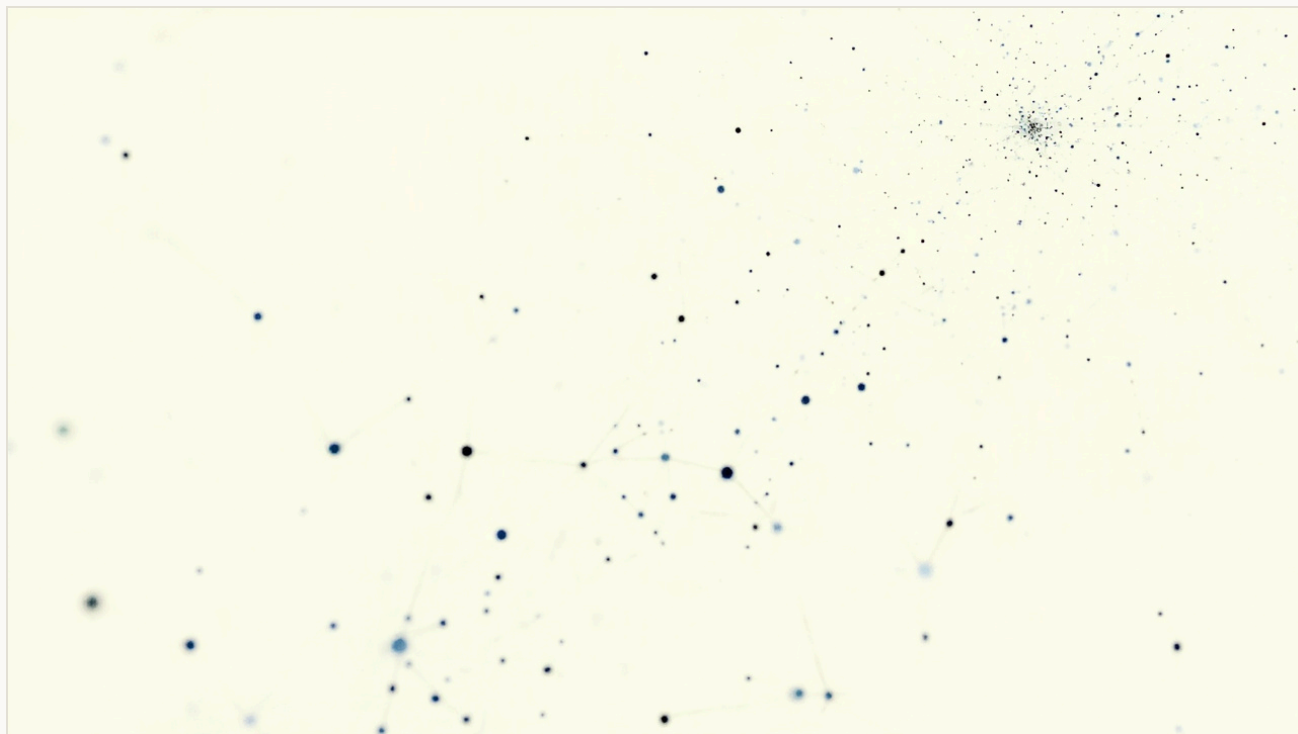
Five Materials That Could Unlock 5–12 GtCO₂/Year

<https://lupine.science/articles/five-materials-for-5-to-12-gtco2-year/>

CLIMATE PARTNERSHIPS PROOF PACK

Type: article **Date:** 2026-07-09 **Scope:** A portfolio-level view of the five material classes Lupine Science is targeting for climate impact. **Description:** Five computationally intractable materials—cobalt-free cathodes, halide solid electrolytes, MOF sorbents, ammonia catalysts, and lead-free perovskites—could unlock 5–12 GtCO₂/year, but only if the predictions sent to labs are corrected and verified first. **Audience:** sophisticated materials, mechanical, and chemical engineers; climate-tech investors **Status:** Draft

Five Materials That Could Unlock 5–12 GtCO₂/Year



Five indigo nodes in a sparse constellation: the material bottlenecks that could unlock 5–12 GtCO₂/year.

The difference between a climate model and a climate outcome is usually a material. Batteries need cathodes that survive a thousand cycles. Solid-state cells need electrolytes that conduct lithium without conducting dendrites. Direct air capture needs sorbents that bind CO₂ at 400

ppm and survive humidity. Ammonia synthesis needs catalysts that split dinitrogen without the Haber-Bosch furnace. Solar needs absorbers that rival lead perovskites without the lead. None of these materials exists at the required scale and price.

Lupine Science is not trying to invent them one at a time. We are building a correction-and-verification layer that makes computational discovery trustworthy enough to screen the multi-component spaces where these materials live. The result is a five-target portfolio with a combined climate potential of **5–12 GtCO₂/year**^[1].

This article explains why those five targets were chosen, what makes each one computationally hard, and how a single methodological idea—measuring systematic error as a physical field over local atomic environments—translates into progress across all of them. The aggregate number is not a fundraising flourish; it is the sum of five independently grounded impact estimates, each contingent on solving a specific defect-chemistry problem.

How the targets were selected

Each target satisfies three filters.

First, **gigatonne-scale climate impact**. The International Energy Agency estimates that batteries alone are directly linked to roughly 20% of the CO₂ reductions required by 2030, and indirectly to another 40%^[2]. Two of our five targets are battery materials for exactly that reason; the other three address industrial emissions, legacy carbon removal, and solar deployment.

Second, **discovery difficulty that has defeated existing methods**. In every case, the property that determines performance—ion migration, defect formation, catalytic barrier, oxidation resistance—occurs at under-coordinated atomic environments where universal machine-learning interatomic potentials (uMLIPs) systematically soften the potential energy surface^[3]. Raw uMLIPs are fast but wrong where it matters; DFT is right but economically impossible at screening scale. The result is a generation of predicted materials that look stable on paper and fail in the synthesis vessel or the device.

Third, a **precise Lupine mechanism**. Each target maps a known failure mode onto a specific correction or proof. LMR cathodes require corrected transition-metal migration barriers. Halide electrolytes require corrected Li^+ hop barriers. MOFs require corrected hydrolysis energies and impossibility proofs for unsynthesizable frameworks. Ammonia catalysts require corrected N_2 dissociation barriers and selective flags for scaling-relation breakers. Lead-free perovskites require corrected Sn vacancy formation energies and provable boundaries for metastable phases.

The targets are not independent scientific bets. They are five instances of the same defect-mediated problem. This is deliberate: a platform that solves one of them credibly has a path to solving all of them, because the correction layer transfers across any material family whose error field can be anchored to measurable observables.

Batteries: cathodes and electrolytes as a pair

Cobalt-free lithium-manganese-rich cathodes

The highest-leverage near-term target is a cobalt-free lithium-manganese-rich (LMR) cathode exceeding **300 Wh/kg cell-level energy density**, cycling stability above **1,000 cycles**, and cost below **\$80/kWh**. The EaCAM consortium at Argonne National Laboratory has demonstrated cobalt-free LMR systems at approximately **270 Wh/kg** and $\sim \$80/\text{kWh}$ ^[4].

The remaining barrier is voltage fade. LMR cathodes derive excess capacity from oxygen redox, but oxygen redox triggers oxygen loss, transition-metal migration from octahedral to tetrahedral sites, and surface reconstruction into spinel or rock-salt phases that block lithium diffusion^[5]. Removing cobalt—essential for supply-chain security, since the Democratic Republic of Congo produces roughly 70% of global cobalt^[6]—exacerbates the instability.

The compositional space is enormous. When concentration gradients, coatings, and doping profiles are included, the candidate count exceeds 10^6 ^[7]. uMLIPs systematically underestimate the migration barriers that govern voltage fade because the transition state involves under-coordinated metal ions (coordination number $c \approx 5-7$) far from the $c=12$ bulk configurations in their training data. The result is candidate rankings that can be wrong by orders of magnitude

in ionic mobility. A **100 meV** barrier error changes the hopping rate by roughly $e^{(E/kBT)} \approx 50\times$ at room temperature, enough to invert the ranking of candidate compositions by ionic mobility^[8].

Lupine's environment error field corrects those barriers at runtime. The correction is smooth over coordination-number space, so chemically similar compositions receive similar corrections and the true best candidate is not buried by ranking inversion. Formal verification then checks that predicted voltage profiles remain ordered after correction, preventing false-positive synthesis attempts.

Earth-abundant halide solid electrolytes

The second battery target is a halide solid electrolyte in the Li-Zr-Cl or Li-Fe-Cl family with ionic conductivity above **10 mS/cm**, electrochemical stability against lithium metal, and moisture tolerance. Solid-state batteries with lithium metal anodes are the most credible path to **>400 Wh/kg** and the elimination of thermal runaway. The market is projected to grow from **\$886 million in 2024 to \$24.3 billion by 2034**^[9].

Current halide electrolytes such as Li_3InCl_6 and Li_3YCl_6 achieve **1–12 mS/cm**^[10], but indium and yttrium are critical raw materials. Replacing them with Zr, Fe, Al, or Mg could cut cost by roughly an order of magnitude while preserving the moisture tolerance that sulfides lack.

The computational challenge is the same shape as the cathode problem, but the property is Li^+ hop barrier rather than transition-metal migration. Raw uMLIPs underestimate migration barriers by **60%+** due to PES softening at under-coordinated transition states^[11]. Because ionic conductivity depends exponentially on barrier— $\sigma \propto e^{(-E_a/kBT)}$ —a 60% error versus a 15% error changes predicted conductivity by roughly **5,000×** at a typical 300 meV barrier and room temperature^[12]. Fast-ion conductors are discarded as insulators before an experimentalist ever sees them.

Lupine corrects barriers on a representative halide such as Li_2ZrCl_6 and transfers the field across the Li-M-Cl space because all members share the same close-packed anion sublattice. Grain-boundary screening, which requires 500+ atom supercells where DFT costs **10^5 CPU-hours per single-point energy**^[13], becomes feasible with corrected uMLIPs at $\sim 10^{-4}$

s/atom/step. The false-negative elimination that plagues raw uMLIP screening is replaced by a ranked list whose top entries have DFT-level barrier accuracy.

Carbon removal: MOFs for direct air capture

Direct air capture is the only technology that can address legacy emissions regardless of source sector. The IPCC AR6 estimates cumulative carbon dioxide removal needs of **100–1,000 GtCO₂ by 2100**, with annual rates approaching **10 GtCO₂/year** by mid-century^[14]. The material bottleneck is the sorbent.

Our target is a metal-organic framework (MOF) with CO₂ working capacity above **2 mmol/g at 400 ppm**, stability under **40–70% relative humidity**, and scalable synthesis cost below **\$50/kg**. Amine-functionalized MOF-808 has reached **1.2 mmol/g** at 400 ppm and 50% relative humidity^[15]. At **1 GtCO₂/year**, each **\$10/tCO₂** cost reduction saves **\$10 billion annually**^[16].

The MOF design space spans trillions of structures, yet only **~10,636 MOFs** are experimentally synthesized versus over **114,000 computationally generated but unsynthesized**^[17]. Stability is the harder problem than capacity. Water competes with CO₂ for binding sites and hydrolyzes metal-linker bonds; predicting humidity-stable frameworks requires evaluating defect formation energies at metal-linker bonds—exactly the under-coordinated environments where uMLIPs are least accurate^[18].

Lupine's correction addresses metal-linker bond dissociation energies, the key determinant of hydrolytic stability. Its impossibility proofs flag frameworks where mixed-metal nodes create coordination environments outside the measured field, separating candidates worthy of experimental investment from computationally unsupported ones. Ranking preservation then keeps the best humidity-stable frameworks at the top of a **10⁵+ framework** screen, rather than scattering them beneath false-positive high-capacity candidates.

Industrial decarbonization: electrochemical ammonia catalysts

Ammonia synthesis via the Haber-Bosch process consumes **1–2% of global energy** and emits more than **450 MtCO₂/year**, primarily through steam methane reforming^[19]. Electrochemical synthesis at ambient conditions, powered by renewable electricity, could eliminate those emissions while enabling distributed fertilizer production.

Lithium-mediated electrochemical nitrogen reduction has achieved **>90% Faradaic efficiency** at ambient conditions, but energy efficiency is stuck near **28%** because lithium plating requires very negative potentials, dissipating more than 70% of input energy as heat^[20]. The U.S. Department of Energy target is **>60% energy efficiency** at current densities above **300 mA/cm²**^[21].

The challenge is N≡N triple-bond activation—dissociation energy **945 kJ/mol**—under conditions where the hydrogen evolution reaction is thermodynamically favored^[22]. DFT screening of N₂ adsorption on stepped surfaces spans coordination environments from $c\approx 4$ to $c\approx 9$, the full range where the error field operates. uMLIPs underestimate these binding energies due to PES softening, generating false-positive catalyst identifications. Scaling relations between N₂ and NH_x binding energies compound the error: a mistake in one binding energy propagates to all others, potentially inverting turnover-frequency rankings^[23].

Lupine corrects N₂ dissociation barriers on under-coordinated active sites, filtering out catalysts that would be experimentally inactive. More subtly, the field's selective failure identifies scaling-relation-breaking catalysts that conventional screening misses. Single-atom or multi-metal sites with cooperative binding violate the first-shell approximation; Lupine's impossibility proofs flag them for higher-level treatment, directing ab initio effort to the small subset most likely to exceed the volcano-peak activity limit.

Solar: lead-free perovskite absorbers

Lead-based halide perovskites have climbed from ~3% efficiency in 2009 to **>26% single-junction** and ~34.6% in perovskite-silicon tandems^[24]. The market is projected at **\$11 billion by 2033**^[25]. But lead toxicity threatens to limit terawatt-scale deployment, and the EU RoHS and REACH frameworks are progressively restricting lead.

The target is a lead-free perovskite absorber with certified power conversion efficiency above **20%** and operational stability above **25 years**. Tin perovskites are the most promising alternative, but Sn^{2+} oxidizes to Sn^{4+} in minutes under ambient conditions, and efficiencies have stalled near **9%** versus $>26\%$ for lead^[26].

The central degradation mechanism is oxygen insertion through Sn vacancy formation. uMLIPs systematically underestimate Sn vacancy formation energies because vacancies create under-coordinated nearest neighbors that bulk-trained models predict as too stable^[27]. Correcting those energies identifies compositions where tin is most strongly bound and most resistant to oxidation, narrowing a vast double-perovskite search space to the subset worth synthesizing.

Metastability adds a second layer of difficulty. Many high-efficiency perovskites are kinetically trapped during solution processing; the equilibrium phase is often a non-perovskite polymorph with poor photovoltaic properties. Standard convex-hull screening discards these metastable phases, eliminating the compositions that achieve the highest efficiencies^[28]. Lupine's impossibility proofs establish boundaries between candidates where predictions are reliable and candidates requiring synthesis-route engineering to access the metastable phase.

Synthesis: one failure mode, five targets

The five targets span batteries, carbon removal, industrial chemicals, and solar, but they share a common structure. In every case, functional performance is determined by defect-mediated properties in multi-component spaces. In every case, raw uMLIPs systematically soften the potential energy surface at the under-coordinated configurations that govern those properties. And in every case, brute-force DFT is too slow to screen the required compositional space.

Lupine's response is not to train a bigger model. It is to measure the systematic error as a physical field over local atomic environments, correct it at runtime with analytic forces, and verify the resulting claims through machine-checked proof. The environment error field achieves **Pearson $r=0.906$** in blind prediction of never-fitted surface energies across 36 (model, material) combinations with zero adjustable parameters^[29]. Runtime correction adds **15.6% overhead** in the current Python implementation and will drop below **1%** in a compiled

LAMMPS overlay^[30]. **77 build-locked Lean 4 theorems with zero sorry proofs** provide guarantees that statistical validation cannot match^[31]. Where correction fails, the platform proves impossibility rather than reporting a p-value, preventing experimental resources from being spent on computationally unsupported candidates.

This is why the targets form a portfolio rather than a list. The same correction layer that makes LMR cathode screening reliable also makes halide electrolyte, MOF, ammonia catalyst, and perovskite screening reliable. The moat deepens with each campaign: every screen adds validated field measurements, every impossibility proof sharpens the boundary of applicability, and every experimental validation tightens the feedback loop.

From predictions to partners

Predictions are necessary but not sufficient. The path from a corrected energy landscape to a commercial material runs through named experimental collaborators who can synthesize, characterize, and scale the top candidates.

For LMR cathodes, the immediate partners are the **Manthiram Laboratory at UT Austin** and **TexPower EV Technologies**, which operates a **15-ton/year pilot facility** producing NMA cathodes at **>230 mAh/g**^[32]; the **Battery500 Consortium** led by Pacific Northwest National Laboratory, with **350 Wh/kg pouch cells demonstrated at >600 cycles**^[33]; and **Forge Nano**, whose atomic-layer-deposition coatings improve cycle life by **30%+**^[34]. For halide electrolytes, the Tier-1 list includes the **CEDER Group at UC Berkeley** and **Lawrence Berkeley National Laboratory**, the **Janek/Zeier Group at the University of Münster**, **Argonne National Laboratory**, and **Solid Power Inc.**^[35].

For MOFs, the starting points are **UC Berkeley's Yaghi and Long groups**, **Northwestern's Farha Group**, and **BASF**, the only commercial-scale MOF producer at **>200 tons/year**^[36]. For ammonia catalysts, **DTU's Chorkendorff group**—which published the landmark Ca-mediated nitrogen reduction result in *Nature Materials* in 2024^[37]—and **Stanford's SUNCAT Center** provide rigorous verification and computational catalysis expertise. For lead-free perovskites, **NREL**, the **University of Queensland's Wang Group**—holder of the certified

16.65% tin-halide record^[38]—and **Tandem PV Inc.** anchor the path from absorber to tandem module.

These partnerships are the subject of the next article in this series. The point here is methodological: a correction layer without experimental partners is an academic exercise, and experimental partners without corrected predictions are flying blind. The 5–12 GtCO₂/year figure is reachable only when the two are coupled, and only if each candidate is validated through the chain of synthesis, characterization, cell or module testing, and scale-up that turns a predicted crystal into a deployable technology.

Footnotes

1. Aggregate range derived from the individual targets in Lupine Science, *Strategic Discovery Plan: High-Impact Materials for Climate*, 2026: 2–5 GtCO₂ avoided (LMR cathodes), 1–3 GtCO₂ avoided (halide SEs), 0.5–2 GtCO₂/year removed (MOF DAC), ~0.45 GtCO₂/year avoided (ammonia catalysts), 0.5–1 GtCO₂/year avoided (lead-free perovskites). ↩
2. International Energy Agency, *Net Zero by 2050 Roadmap*, 2023. ↩
3. Deng et al., 2024; see Lupine Science, *Strategic Discovery Plan*, §2.1, for the survey of uMLIP errors across bulk formation energies, surface energies, defect formation energies, phonon frequencies, and migration barriers. ↩
4. EaCAM consortium, Argonne National Laboratory, as cited in Lupine Science, *Strategic Discovery Plan*, §4.1.1. ↩
5. Lupine Science, *Strategic Discovery Plan*, §4.1.2. ↩
6. IEA critical minerals analysis and Lupine Science, *Strategic Discovery Plan*, §5.2. ↩
7. Lupine Science, *Strategic Discovery Plan*, §4.1.2. ↩
8. Lupine Science, *Strategic Discovery Plan*, §4.1.3. ↩

9. Lupine Science, *Strategic Discovery Plan*, §4.2.1; market projection source cited therein. ↩
10. Lupine Science, *Strategic Discovery Plan*, §4.2.1. ↩
11. Deng et al., 2024, as cited in Lupine Science, *Strategic Discovery Plan*, §4.2.2. ↩
12. Lupine Science, *Strategic Discovery Plan*, §4.2.3. ↩
13. Lupine Science, *Strategic Discovery Plan*, §4.2.2. ↩
14. IPCC, *AR6 Working Group III*, 2022, as cited in Lupine Science, *Strategic Discovery Plan*, §4.3.1. ↩
15. Lupine Science, *Strategic Discovery Plan*, §4.3.1. ↩
16. Lupine Science, *Strategic Discovery Plan*, §4.3.1. ↩
17. Lupine Science, *Strategic Discovery Plan*, §4.3.2. ↩
18. Lupine Science, *Strategic Discovery Plan*, §4.3.2. ↩
19. Lupine Science, *Strategic Discovery Plan*, §4.4.1. ↩
20. Lupine Science, *Strategic Discovery Plan*, §4.4.1. ↩
21. U.S. Department of Energy, ARPA-E REFUEL program target, as cited in Lupine Science, *Strategic Discovery Plan*, §4.4.1. ↩
22. Lupine Science, *Strategic Discovery Plan*, §4.4.2. ↩
23. Lupine Science, *Strategic Discovery Plan*, §4.4.2. ↩
24. Lupine Science, *Strategic Discovery Plan*, §4.5.1. ↩
25. Lupine Science, *Strategic Discovery Plan*, §4.5.1. ↩
26. Lupine Science, *Strategic Discovery Plan*, §4.5.1. ↩
27. Lupine Science, *Strategic Discovery Plan*, §4.5.3. ↩
28. Lupine Science, *Strategic Discovery Plan*, §4.5.2. ↩
29. Lupine Science, *Strategic Discovery Plan*, §Executive Summary and §3. ↩
30. Lupine Science, *Strategic Discovery Plan*, §Executive Summary and §5.1. ↩

31. Lupine Science, *Strategic Discovery Plan*, §Executive Summary. ↩
32. Lupine Science, *Strategic Partnership Mapping Document*, §1, “Manthiram Laboratory / UT Austin & TexPower EV Technologies.” ↩
33. Lupine Science, *Strategic Partnership Mapping Document*, §1, “Battery500 Consortium (PNNL-led).” ↩
34. Lupine Science, *Strategic Partnership Mapping Document*, §1, “Forge Nano, Inc.” ↩
35. Lupine Science, *Strategic Partnership Mapping Document*, §2. ↩
36. Lupine Science, *Strategic Partnership Mapping Document*, §3. ↩
37. Lupine Science, *Strategic Partnership Mapping Document*, §4, “Technical University of Denmark – Villum Center for Sustainable Fuels (Chorkendorff).” ↩
38. Lupine Science, *Strategic Partnership Mapping Document*, §5, “University of Queensland / Wang Group.” ↩



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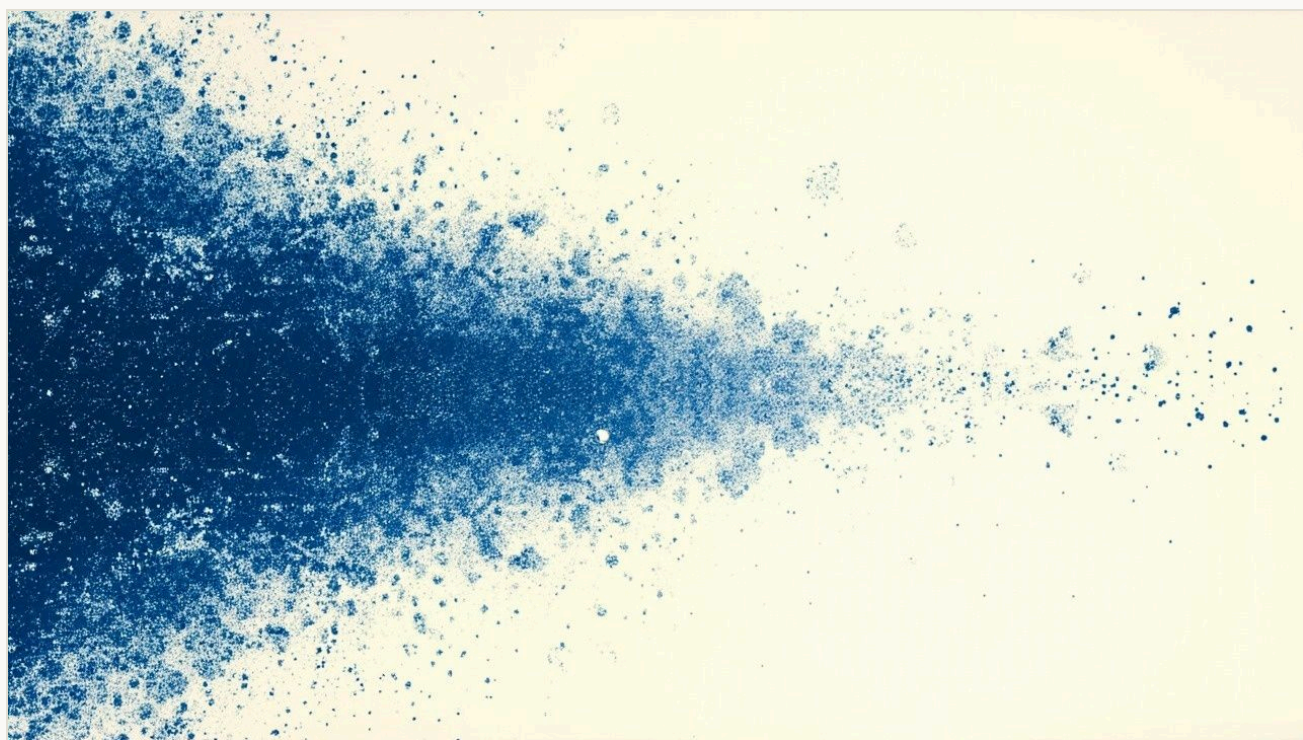
From Predicted Crystal to Commercial Cell

<https://lupine.science/articles/from-predicted-crystal-to-commercial-cell/>

CLIMATE PARTNERSHIPS PROOF PACK

Type: article **Date:** 2026-07-09 **Scope:** How Lupine Science turns computational predictions into commercial climate materials through a sequenced partner chain. **Description:** A rigorous, partner-by-partner map from predicted crystal to commercial cell for batteries, carbon removal, ammonia, and solar absorbers. **Audience:** sophisticated materials, mechanical, and chemical engineers; climate-tech investors **Status:** Draft

From Predicted Crystal to Commercial Cell



From bits to atoms: a bold indigo ribbon crossing from abstract structure toward makeable hardware.

A predicted crystal is not a material. A synthesized powder is not a cell. A cell is not a fleet, a filter, a fertilizer plant, or a solar farm. Between a computational screen and climate-relevant deployment stands a chain of specialized capabilities – synthesis, coating, cell fabrication, operando characterization, module integration, and manufacturing scale-up – that no single

institution owns. The question is not simply “which compound is stable?” but “which compound can survive handoff across this chain?”

The conventional answer is to build a bigger generator. That has produced volume without trust. Google DeepMind’s GNoME predicted 2.2 million crystals and identified 380,000 as thermodynamically stable, yet only 736 had been independently synthesized by late 2023 – a 0.2% validation rate^[1]. The A-Lab autonomous synthesis facility reported 63% success, but independent critique found that two-thirds of its “novel” targets were already-known disordered phases, reducing the true novel discovery rate to near zero^[2]. The bottleneck is not a shortage of predictions; it is the absence of a reliable correction and verification layer between prediction and experiment.

Lupine Science occupies that layer. Rather than competing with structure generators, laboratories, or databases, Lupine corrects the systematic errors of universal machine-learning interatomic potentials (uMLIPs) using a measured environment error field, deploys the correction at runtime with analytic forces, and verifies claims through machine-checked Lean 4 proofs^[3]. The result is a partner-facing platform that makes other organizations’ predictions trustworthy, accelerating every link from crystal to cell.

This article maps that chain across Lupine’s five priority targets – cobalt-free lithium-manganese-rich (LMR) cathodes, earth-abundant halide solid electrolytes, metal-organic frameworks (MOFs) for direct air capture, electrochemical ammonia catalysts, and lead-free perovskite solar absorbers. In each, a prediction is handed to synthesis, then characterization, then integration, then manufacturing. Where each handoff currently fails, a correction-and-verification layer creates value.

Batteries: From Cathode Prediction to Pack Integration

LMR cathodes are the highest-impact near-term target. By eliminating cobalt and raising energy density above 300 Wh/kg, they could avoid 2–5 GtCO₂ cumulatively by 2050^[4]. The commercial stakes are visible in the field: POSCO Future M has completed pilot LMR production and plans mass manufacturing within 2025, while GM has announced commercial LMR deployment by 2028^[5].

The first handoff is from computation to synthesis. Lupine's corrected migration barriers and oxygen-vacancy formation energies rank candidates across a compositional space exceeding 10^6 compositions^[4:1]. The Manthiram Laboratory at UT Austin is the natural Tier 1 partner: Arumugam Manthiram invented cobalt-free layered oxide cathodes and has trained more than 300 researchers^[5:1]. TexPower EV Technologies – a Manthiram spinout – extends this to scale-up with a 15-ton-per-year pilot producing NMA cathodes at >230 mAh/g and a 300-ton plant planned by 2027^[5:2].

The second handoff is from powder to coated particle. Voltage fade is driven by transition-metal migration and surface reconstruction, both suppressible by atomic-layer-deposition (ALD) coatings. Forge Nano operates 15 ALD systems, processes from 1 g to 30,000 kg per day, and has demonstrated that ALD coatings improve cycle life by more than 30% while reducing resistance five-fold^[5:3]. A three-way arrangement – Lupine identifies compositions, Forge Nano applies coatings, TexPower supplies coated material – closes the cathode-to-cell gap.

The third handoff is from material to cell. The Battery500 Consortium, led by Pacific Northwest National Laboratory and spanning four national labs and six universities, operates pouch-cell fabrication lines and has demonstrated 350 Wh/kg pouch cells with more than 600 cycles^[5:4]. Lupine fits as a computational seedling project feeding the consortium's cathode pipeline.

Finally, the OEM handoff. GM's Wallace Battery Cell Innovation Center has coated 1 ton of LMR cathode material, built 18 prototype cells, and logged 1.4 million equivalent miles of testing^[5:5]. LG Energy Solution holds the world's largest LMR patent portfolio and is commercializing prismatic cells with GM for 2028^[5:6]. These industrial partners do not need another screen; they need a screen they can trust.

Batteries: From Halide Electrolyte to Solid-State Cell

If LMR cathodes are the near-term lever, solid-state batteries with lithium-metal anodes are the long-term leap. Earth-abundant halide solid electrolytes – Li-Zr-Cl and Li-Fe-Cl systems –

target >10 mS/cm ionic conductivity, stability versus lithium metal, and compatibility with mechanochemical synthesis^[6]. The solid-state battery market is projected to grow from \$886 million in 2024 to \$24.3 billion by 2034^[4:2].

The discovery challenge is that raw uMLIPs systematically underestimate Li^+ migration barriers by 60% or more because transition states are under-coordinated relative to bulk training data^[4:3]. A 60% barrier error changes predicted room-temperature conductivity by roughly 5,000×. Lupine's environment error field corrects this at uMLIP speed.

The partner chain begins at UC Berkeley with Gerbrand Ceder's group and the Materials Project, which published foundational DFT studies of Li_3YCl_6 and Li_2ZrCl_6 ^[6:1]. The University of Münster's Janek and Zeier groups add world-leading halide solid-electrolyte synthesis and lithium-metal interface analytics via XPS and cryo-TEM^[6:2]. Argonne National Laboratory contributes operando X-ray absorption spectroscopy and the Materials Engineering Research Facility for 20–100 kg scale-up^[6:3].

Industrial translation runs through Solid Power, which produces sulfide and halide solid electrolytes, operates roll-to-roll manufacturing, and has built 100+ Ah cells validated by BMW^[6:4]. Factorial Energy, whose FEST/Solstice electrolytes have achieved 391 Wh/kg cells with Mercedes EQS validation, offers another integration path^[6:5]. Ion Storage Systems, with a \$20 million ARPA-E SCALEUP award and a Saint-Gobain partnership, provides an anodeless-manufacturing route that could pair with lower-cost halide chemistry^[6:6].

The cross-cutting partner here is again Battery500. Lupine's halide-screening results and the consortium's pouch-cell fabrication create a single loop: computational composition → mechanochemical synthesis → impedance spectroscopy → full-cell cycling. With correction, each handoff carries a bounded error budget.

Carbon Removal: From MOF Linker to Gigafactory

Direct air capture is the only carbon-removal technology that can address legacy emissions regardless of source sector. The IPCC estimates cumulative removal needs of 100–1,000 GtCO₂ by 2100, with annual rates approaching 10 GtCO₂/year by mid-century^[4:4]. MOFs are

the most tunable sorbent class, but the gap between a predicted stable framework and a deployed filter is wide: synthesis costs must fall below \$50/kg, humidity stability must survive thousands of cycles, and working capacity must exceed 2 mmol/g at 400 ppm CO₂ ^[6:7].

The partner chain starts with reticular chemistry. UC Berkeley's Omar Yaghi and Jeffrey Long offer synthesis of >100,000 MOF/COF variants and cooperative-binding design ^[6:8]. Omar Farha's group at Northwestern brings MOF-808 and Zr-based synthesis expertise, plus a \$3.3 million DOE program focused on DAC MOF stability ^[6:9].

Lupine's role is to correct hydrolysis-barrier errors and flag unsupported frameworks. Water attack on metal-linker bonds proceeds through under-coordinated transition states – exactly where uMLIPs soften the potential energy surface ^[4:5]. Corrected hydrolysis barriers filter frameworks by real-world humidity stability, not idealized gas-phase energy.

Manufacturing scale runs through BASF, the first commercial MOF producer at >200 tons per year and exclusive manufacturer of Svante's CALF-20 sorbent ^[6:10]. Svante has deployed MOF-based capture at 25 tonnes per day and is planning a 10 million tCO₂/year “gigafactory” ^[6:11]. Climeworks, the world's largest DAC deployer with its Mammoth plant in Iceland, offers field validation ^[6:12]. Atoco, Yaghi's commercialization spinout, provides a startup route to co-development ^[6:13].

Industrial Decarbonization: From Catalyst Prediction to Ammonia Market

Haber-Bosch ammonia synthesis emits approximately 450 MtCO₂/year and consumes 1–2% of global energy ^[4:6]. Electrochemical synthesis powered by renewable electricity could eliminate those emissions. Lithium-mediated nitrogen reduction has achieved >90% Faradaic efficiency, but energy efficiency is stuck near 28% because lithium plating imposes a >3 V overpotential penalty ^[4:7]. Lupine targets non-lithium mediators – Ca, Mg, Al, Na – with a goal of >60% energy efficiency and >300 mA/cm² partial current density ^[6:14].

The most rigorous verification partner is the Technical University of Denmark's Villum Center for Sustainable Fuels, led by Ib Chorkendorff. DTU published the landmark Ca-mediated

nitrogen reduction result in 2024, achieving 40% Faradaic efficiency, and operates the world's most rigorous ammonia verification protocol^[6:15]. Chorkendorff has described non-lithium mediators as the field's highest priority.

Stanford's SUNCAT Center, co-directed by Jens Nørskov, adds computational catalysis and operando X-ray absorption spectroscopy at SLAC^[6:16]. The loop is tight: Lupine predicts compositions, SUNCAT screens them at the electronic-structure level, and DTU validates the winners.

Commercial engineering comes through Jupiter Ionics, which licensed Douglas MacFarlane's lithium-mediated technology and is developing modular MSA Cell technology after a \$9 million Series A^[6:17]. ARPA-E's REFUEL program provides funding and metrics: >86% energy efficiency, >300 mA/cm² at 90% current efficiency, and >100 g/day electrochemical ammonia^[6:18].

Downstream, Caltech's Karthish Manthiram group contributes gas-diffusion-electrode engineering, Nitricity offers distributed-fertilizer piloting at its Delhi, California plant, and CF Industries – the world's largest ammonia manufacturer at ~10 million tons per year – provides the eventual industrial exit^[6:19].

Solar: From Lead-Free Absorber to Tandem Module

Lead-based halide perovskites have exceeded 26% single-junction efficiency and ~34.6% in perovskite-silicon tandems, but lead toxicity and impending EU restrictions threaten terawatt-scale deployment^[4:8]. Lead-free tin perovskites are the leading alternative, yet Sn²⁺ oxidizes within minutes to hours and efficiencies have stalled near 9%^[4:9]. The target is >20% certified power conversion efficiency with >25-year operational stability^[6:20].

The key descriptor is Sn vacancy formation energy, which determines oxidation susceptibility. uMLIPs systematically underestimate this energy because vacancies create under-coordinated Sn neighbors that bulk-trained models predict as too stable^[4:10]. Lupine's correction recovers accurate vacancy energetics, while provable boundaries flag metastable phases that convex-hull screening would discard.

The partner chain begins with NREL, which provides certified efficiency measurements, slot-die and roll-to-roll fabrication, and ISOS stability protocols^[6:21]. The University of Queensland's Lianzhou Wang group holds the certified world record for tin halide perovskites at 16.65% (April 2025) and has demonstrated 1,500-hour stability without encapsulation^[6:22]. Tandem PV Inc., which has raised \$87 million and operates a pilot factory in San Jose, integrates perovskite top cells with silicon bottom cells^[6:23]. Northwestern's Mercuri Kanatzidis, who pioneered CsSnI₃ and solid-state halide perovskite solar cells, offers synthesis and panoramic methodology for discovering new absorber families^[6:24].

The scale-up handoff runs through Oxford Photovoltaics, which began commercial tandem shipments in 2024 and holds a 26.9% world-record module efficiency, and through Swift Solar, whose vapor-deposition manufacturing is especially compatible with oxygen-sensitive tin perovskites^[6:25].

Cross-Cutting Architecture: NREL, UC Berkeley, and ARPA-E

Each target has its own partner chain, but three institutions create economies of scope.

NREL is the highest-value cross-cutting partner. It spans four of Lupine's five targets – cathodes, halide solid electrolytes, ammonia catalysts, and perovskites – and offers certified testing, lifecycle analysis, techno-economic analysis, and a single point of access to DOE funding networks^[6:26]. A master CRADA with NREL replaces multiple one-off agreements.

UC Berkeley offers a complementary two-group engagement: the Ceder group for battery materials and the Yaghi/Long groups for MOF direct air capture^[6:27]. A campus-wide partnership gives Lupine access to synthesis, characterization, and autonomous discovery infrastructure across three targets from one institutional relationship.

ARPA-E should be approached with a portfolio strategy rather than individual proposals. Coordinated submissions across REFUEL (ammonia), IONICS (halide solid electrolytes), and OPEN (MOF DAC) position Lupine as a multi-target computational materials company^[6:28]. The return is non-dilutive capital and validation credibility. ARPA-E's \$3.5 billion portfolio has catalyzed \$11.8 billion in private follow-on funding^[7], and the NIST Materials Genome

Initiative analysis estimates \$123 billion–\$270 billion in annual value from improved materials innovation infrastructure^[7:1].

The Verification Layer as Partnership Enabler

Partners validate the science; verification makes it trustworthy at scale. Every organization named above has been burned by computational predictions that looked convincing in a paper but failed in the lab. The 0.2% GNoME synthesis rate and the A-Lab disordered-phase episode are warnings that experimental budgets are finite and false positives are expensive.

Lupine's environment error field, runtime correction, and formal verification address this directly. The field achieves $r=0.906$ blind prediction of never-fitted surface energies across 36 (model, material) combinations with zero adjustable parameters^[3:1]. Runtime correction adds 15.6% overhead in Python and will drop below 1% in a compiled LAMMPS overlay, keeping corrected uMLIPs roughly $10^5\times$ faster than DFT^[3:2]. Seventy-seven build-locked Lean 4 theorems provide machine-checked guarantees^[3:3]. Where correction cannot apply, Lupine proves impossibility rather than reporting a p-value.

For partners, handoffs carry documentation. A cathode candidate comes with corrected migration barriers and a theorem certificate of the ranking claim. A MOF candidate comes with a hydrolysis-barrier correction and a proof of whether its coordination environment lies inside or outside the field's domain. An ammonia catalyst comes with corrected N_2 dissociation barriers and a flag if scaling-relation breaking requires higher-level treatment. Experimentalists still synthesize and measure; Lupine removes the guesswork about which candidates deserve the work.

In a discovery ecosystem that must scale from thousands to millions of validated materials per year, the scarce resource is not prediction volume. It is the trust that turns a predicted crystal into a commercial cell.

Footnotes

1. Google DeepMind’s GNoME predicted 2.2 million crystals, of which 380,000 were computed stable; only 736 had been independently synthesized by late 2023, a 0.2% validation rate. See *Lupine Science Strategic Discovery Plan: High-Impact Materials for Climate*, Section 2.1. ↩
2. The A-Lab autonomous synthesis facility reported 41 novel compounds from 58 targets (63% success), but independent critique found two-thirds of targets were ordered approximations of known disordered phases, reducing the true novel discovery rate to near zero. See *Lupine Science Strategic Discovery Plan*, Section 2.1. ↩
3. Lupine measures systematic error as a physical field over local atomic environments, corrects it at runtime with analytic forces, and verifies claims through 77 build-locked Lean 4 theorems with zero sorry proofs. Blind prediction achieves Pearson $r=0.906$ ($p=10^{-4}$, 95% CI [0.82, 0.96]) across 36 (model, material) combinations with zero adjustable parameters. Runtime correction adds 15.6% overhead in Python and will drop below 1% in a compiled LAMMPS overlay. See *Lupine Science Strategic Discovery Plan*, Section 3. ↩ ↩ ↩ ↩
4. The five priority targets – cobalt-free LMR cathodes, earth-abundant halide solid electrolytes, MOFs for direct air capture, electrochemical ammonia catalysts, and lead-free perovskites – collectively address 5–12 GtCO₂/year. See *Lupine Science Strategic Discovery Plan*, Sections 4.1–4.5 and 6.1.1. ↩ ↩ ↩ ↩ ↩ ↩ ↩ ↩ ↩ ↩ ↩
5. Partnership details for cobalt-free LMR cathodes: Manthiram Laboratory / UT Austin and TexPower EV Technologies; Battery500 Consortium (PNNL-led); Forge Nano, Inc.; POSCO Future M; General Motors / Ultium Cells LLC; LG Energy Solution; Argonne National Laboratory / Advanced Photon Source. TexPower operates a 15-ton/year pilot producing NMA cathodes at >230 mAh/g; GM targets 2028 LMR deployment. See *Lupine Science: Strategic Partnership Mapping Document*, Section 1. ↩ ↩ ↩ ↩ ↩ ↩ ↩
6. Partnership details across all five targets, including tier rankings and key metrics, are from *Lupine Science: Strategic Partnership Mapping Document*, Sections 1–5 and Cross-Cutting Partners Table. ↩
7. ARPA-E’s \$3.5 billion portfolio (2009–2023) catalyzed \$11.8 billion in private follow-on funding; the NIST/RTI Materials Genome Initiative economic analysis estimates \$123 billion–\$270 billion in annual value from improved materials innovation infrastructure. See *Lupine Science Strategic Discovery Plan*, Sections 5.4 and 6.1.2. ↩ ↩



Lupine Science

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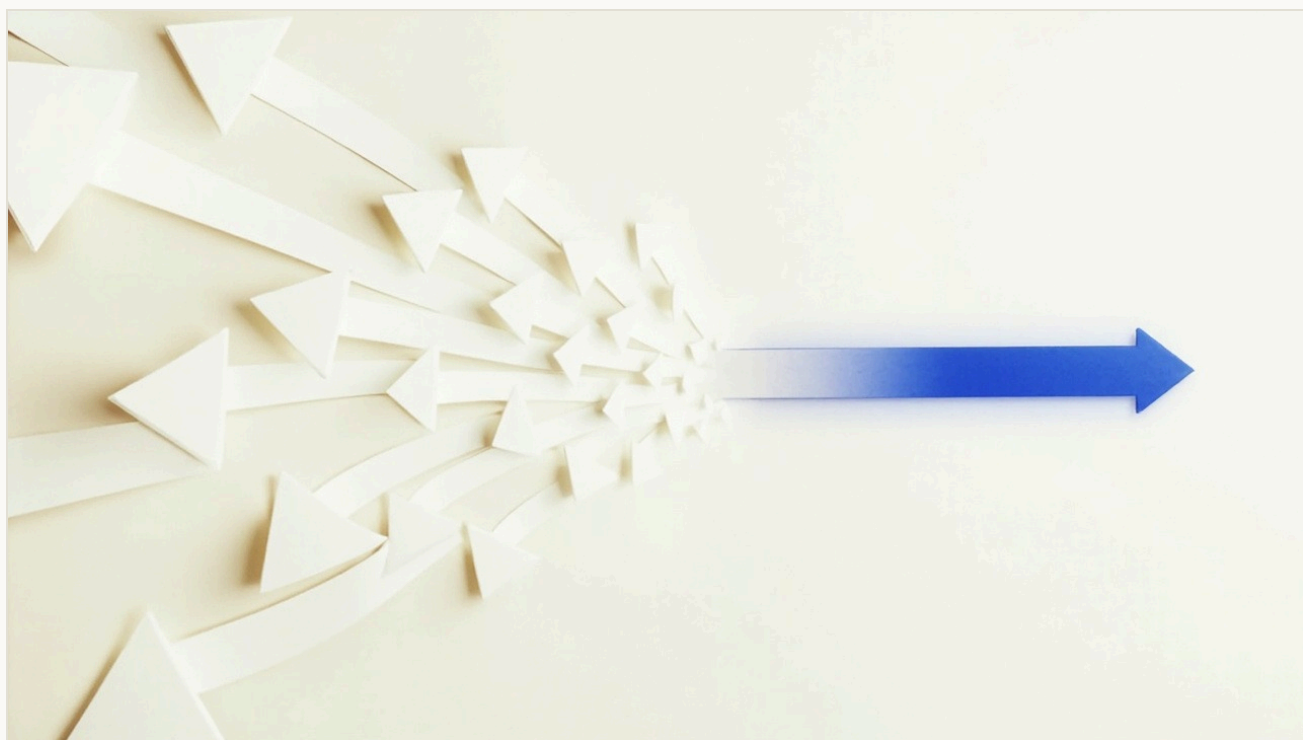
Investing in the Trust Layer

<https://lupine.science/articles/investing-in-the-trust-layer/>

CLIMATE PARTNERSHIPS PROOF PACK

Type: article **Date:** 2026-07-09 **Scope:** Lupine Science's correction-and-verification layer as investable climate infrastructure **Description:** Why trust, not raw prediction volume, is the binding constraint in computational materials discovery, and how Lupine builds it. **Audience:** sophisticated materials, mechanical, and chemical engineers; climate-tech investors **Status:** Draft

Investing in the Trust Layer



Near-parallel error vectors collapsing into alignment: the trust layer for AI-designed matter.

Computational materials discovery is no longer short on imagination. Structure generators such as Google DeepMind's GNoME have proposed 2.2 million candidate crystals; universal machine-learned interatomic potentials (uMLIPs) such as CHGNet and MACE evaluate them at roughly 10^{-4} seconds per atom-step; and autonomous labs such as the A-Lab at Lawrence

Berkeley National Laboratory attempt synthesis around the clock.^[1] The scarce resource is no longer the ability to invent or simulate. It is the ability to believe the result.

Belief is scarce because the pipeline is lossy. GNoME reported 380,000 computed-stable structures, yet only 736 had been independently synthesized by late 2023 – a 0.2% validation rate.^[2] A-Lab reported a 63% experimental success rate, but independent critique found that two-thirds of its “novel” targets were known disordered phases misclassified by the generator; the true novel-discovery rate may have been near zero.^[3] These are not failures of ambition. They are structural symptoms of a field that generates faster than it verifies.

For climate materials, the cost of that asymmetry is measured in gigatonnes and years. Batteries alone are directly linked to roughly 20% of the CO₂ reductions required by 2030 and indirectly to another 40%.^[4] Each false positive sends a team into weeks of wasted synthesis; each false negative buries a candidate that might have reached a cell test. Speed without accuracy wastes experiments; accuracy without speed misses the deployment window. The winning infrastructure is the layer that de-risks the lab queue.

Lupine Science is building that layer.

The layered pipeline, and where it leaks

Computational discovery can be drawn as four stages: generation, prediction, synthesis, and validation. Each stage has dominant platforms; each produces errors that propagate downstream.

At the generation stage, GNoME and Microsoft’s MatterGen produce candidate structures at machine speed. MatterGen reports 38% stable-unique-novel generation rates, but its published “discovery” of TaCr₂O₆ was later identified as a known disordered phase, and its outputs are limited to 20 atoms per unit cell with no treatment of disorder.^[5] GNoME’s 2.2 million predictions excluded polymers, glasses, metal-organic frameworks, heterostructures, and composites – the very material classes most relevant to climate technology.^[6]

At the prediction stage, uMLIPs supply near-DFT energies for the generated structures. They are accurate on bulk properties but systematically soften the potential energy surface away from equilibrium. Across 21 materials and up to nine properties per model, defect-family observables – surfaces, vacancies, stacking faults – err 15–60× worse than bulk observables.^[7] The error is not random: it is a smooth function of local atomic environment, concentrated where coordination deviates from the bulk.

At the synthesis stage, A-Lab and manual labs attempt to make what upstream recommended. Without a characterization of each candidate's failure modes, the queue is filtered by intuition and compute budget rather than by proof. The result is the 0.2%–63% pattern: high activity, low validated throughput.

Lupine does not compete with any of these layers. It sits between prediction and synthesis as a correction-and-verification layer: it measures the systematic error of the predictor, corrects it at runtime, and proves which claims can be trusted. The value proposition is partnership, not displacement. MatterGen and GNoME generate more candidates as they improve; Lupine's market grows because every additional candidate must be checked.

Three pillars of defensibility

The trust layer is defensible only if it is simultaneously accurate, provable, and deployable. Lupine's architecture rests on three pillars that are individually difficult and, to our knowledge, collectively unique in materials science.

Measured error field, not learned. Rather than training a delta-ML model to fit discrepancies per system, Lupine treats the uMLIP error as a physical field over local atomic environments. For face-centered-cubic metals, the field is a cubic spline over first-shell coordination number, fixed by three anchor observables – γ_{100} , γ_{111} , and the vacancy formation energy – with the bulk value constrained to zero at coordination 12. A fourth observable, γ_{110} , is held back as a blind test. Across 36 independent (model, material) combinations, the field predicts the never-fitted γ_{110} error with Pearson $r = 0.906$ (material-clustered 95% CI [0.82, 0.96], $p = 10^{-4}$) and zero adjustable parameters.^[8]

The distinction from delta-ML and fine-tuning matters commercially. Delta-ML requires retraining for every new material, negating the speed advantage of universal potentials. Fine-tuning requires curated target-system data that does not exist for unexplored composition spaces. Lupine’s correction is measured from three standard observables and transfers within a structure family; no per-system training is required. The improvement is concrete: on the Ni(110) blind facet, relative error falls from 9.7% to 1.5%; on Cu(110), from 28.0% to 13.7%.
[9]

Machine-checked proof. Lupine’s quantitative claims are sealed as Lean 4 theorems over integer-scaled data carrying SHA-256 provenance. The formalization currently comprises 8 modules, 77 build-locked theorems, approximately 225 declarations, and zero `sorry` proofs – Lean’s escape hatch for unproven claims.^[10] The theorems cover ordering inequalities, isotonic correction bounds, and impossibility results where no monotone correction can recover the reference ranking.

This is a categorical difference from statistical validation. Google DeepMind and Microsoft can scale compute indefinitely, but they cannot produce machine-checked proofs of why a prediction should be believed. The Lean kernel once rejected a claim that had survived statistical filtering, reducing a reported success count from 27/36 to 26/36 at integer precision.^[11] That episode is the product in miniature: a trust layer must be able to say no to itself.

Runtime compatibility. The correction term is shaped like an EAM embedding function – a sum over atoms of a function of local coordination – so it deploys as a LAMMPS overlay pair style beside a live CHGNet or MACE calculator. Measured wall-time overhead in Python is 15.6%; a compiled C/C++ or CUDA implementation is expected to drop this below 1%.^[12] Even at 15.6%, corrected uMLIPs remain approximately $10^5\times$ faster than DFT for structurally complex cells.^[13] Users keep their existing simulators; Lupine corrects them without retraining.

Why the moat deepens with use

Unlike data-scale moats that erode as competitors scrape the same training corpora, Lupine's position strengthens with each screening campaign. Every measured error field adds to a reference database of systematic biases organized by (model, material, structure family). Each impossibility proof sharpens the boundary of applicability. Each experimental validation tightens the feedback loop between computation and synthesis.

Transfer is the mechanism. The Li-Zr-Cl field informs Li-Fe-Cl halide solid-electrolyte screening; the fcc field informs body-centered-cubic and hexagonal-close-packed extensions planned for Phase 0; the cobalt-free cathode field transfers to layered oxide and spinel dopant spaces.^[14] Theorem families expand in parallel: extending the formalization from fcc to bcc, hcp, and layered structures adds machine-checked protections for the material classes that dominate battery and catalyst targets.

The partnership architecture reinforces the flywheel. A master CRADA with NREL, which spans four of Lupine's five priority targets, would provide certified testing infrastructure and a single point of access to DOE funding networks.^[15] Tier-1 experimental collaborators – the Manthiram Laboratory and TexPower for cobalt-free cathodes, the Ceder group at UC Berkeley and LBNL for halide electrolytes, the University of Münster for Li-metal interface analytics, DTU's Chorkendorff group for ammonia verification, and the Yaghi and Long groups at UC Berkeley for MOF direct air capture – supply the synthesis and characterization data that close the loop.^[16] Each validated measurement makes the next prediction more trustworthy.

The economic case

The macroeconomic argument is not speculative. The NIST Materials Genome Initiative economic analysis estimates that improved materials innovation infrastructure would deliver \$123 billion–\$270 billion in annual value to U.S. industry.^[17] ARPA-E's \$3.5 billion portfolio from 2009 to 2023 catalyzed \$11.8 billion in private follow-on funding and \$21.9 billion in IPO or acquisition value.^[18] Self-driving laboratories funded by ARPA-E (\$40 million) and CHIPS Act investments (\$100 million) could compress materials-development timelines from 10–20 years to 2–5 years.^[19]

Against that backdrop, Lupine's cumulative 36-month budget is approximately \$3.2 million.

^[20] The leverage ratio justifies the expenditure even if only one of the five priority targets – cobalt-free cathodes, halide solid electrolytes, MOF sorbents, electrochemical ammonia catalysts, or lead-free perovskites – yields a commercially viable material. Each target addresses a market projected to exceed \$10 billion by the mid-2030s, and the combined climate impact potential is 5–12 GtCO₂/year. ^[21]

The investment thesis is therefore not a bet on a single material. It is a bet on the infrastructure that makes every material-discovery program more capital-efficient. Corrected screening campaigns become economically possible where brute-force DFT is not. Verified predictions reduce the false-positive tax that currently consumes lab time. Partners across the stack – generators, databases, autonomous labs, OEMs, national labs – capture more value from their own data because the trust layer exists.

Risk and honesty

A proof-first voice requires naming what is still open. The fcc error field is established; extension to bcc, hcp, and layered structures is the critical path item in Phase 0, with a go/no-go gate at Month 3. ^[22] A second-shell correction may be needed for systems where first-shell coordination alone does not resolve the error geometry. Industrial validation partners are targeted but not yet contracted; until synthesis data arrive, the transfer claims remain computational.

These are engineering risks, not conceptual ones. The conceptual question – whether uMLIP errors are structured and measurable – has been answered affirmatively for fcc metals. The remaining work is to extend the field's jurisdiction and to prove, with each new structure family, where correction applies and where it does not.

The trust layer for a real-world Replicator

The long-term vision is a discovery pipeline in which AI-designed matter can be made, measured, and believed. Structure generators propose; Lupine corrects and verifies; experimental partners synthesize; the result feeds back into the field database and the theorem library. The loop accelerates not because errors are hidden but because they are characterized.

The climate window is unforgiving. A material discovered in 2035 may miss the 2040 deployment window, and the cumulative emissions difference between deployment in 2035 versus 2045 is measured in tens of gigatonnes.^[23] The world does not need more predicted crystals. It needs a smaller, defensible set of predictions that laboratories can act on.

That is the trust layer. It is the infrastructure that converts the abundance of generated materials into the scarcity of validated ones – and it is the investment that makes the rest of the climate-materials pipeline economically rational.

Footnotes

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1. See Lupine Science, *Strategic Discovery Plan: High-Impact Materials for Climate*, §1, for the pipeline map and player taxonomy. ↩
 2. Merchant et al., “Scaling deep learning for materials discovery,” *Nature*, 2023; Lupine analysis in *Strategic Discovery Plan*, Executive Summary. ↩
 3. Lupine Science, *Strategic Discovery Plan*, §2.4 and §3.4, citing independent critique of A-Lab’s “novel” target classification. ↩
 4. International Energy Agency, as cited in Lupine Science, *Strategic Discovery Plan*, §1. ↩
 5. Lupine Science, *Strategic Discovery Plan*, §2.3. ↩
 6. Lupine Science, *Strategic Discovery Plan*, §2.3. ↩
 7. Lupine Science, *Strategic Discovery Plan*, §2.2; benchmark corpus covers 21 materials × 4 uMLIPs × up to 9 properties. ↩

8. Lupine Science, *Strategic Discovery Plan*, §3.2; full benchmark and confidence-interval methodology described in the environment-error-field proof pack. ↩
9. Lupine Science, *Strategic Discovery Plan*, §3.2.2 and Figure 2. ↩
10. Lupine Science, *Strategic Discovery Plan*, §3.3.1. ↩
11. Lupine Science, *Strategic Discovery Plan*, §3.3.2, “The Kernel-Rejected Claim.” ↩
12. Lupine Science, *Strategic Discovery Plan*, §3.2.2 and §5.1. ↩
13. Lupine Science, *Strategic Discovery Plan*, §3.4.2, comparing DFT cost for a 500-atom high-entropy-alloy supercell. ↩
14. Lupine Science, *Strategic Discovery Plan*, §4 and §6.3.2. ↩
15. Lupine Science, *Strategic Partnership Mapping Document*, §6 (Cross-Cutting Partners Table). ↩
16. Lupine Science, *Strategic Partnership Mapping Document*, Sections 1–5 (Tier-1 line items for each target). ↩
17. NIST/RTI International Materials Genome Initiative economic analysis, as cited in Lupine Science, *Strategic Discovery Plan*, §6.1.2. ↩
18. ARPA-E portfolio data, 2009–2023, as cited in Lupine Science, *Strategic Discovery Plan*, §6.1.2. ↩
19. Lupine Science, *Strategic Discovery Plan*, §6.1.2. ↩
20. Lupine Science, *Strategic Discovery Plan*, §5 and Table 1. ↩
21. Lupine Science, *Strategic Discovery Plan*, §6.1.1 and §4. ↩
22. Lupine Science, *Strategic Discovery Plan*, §5.1 (Phase 0: Foundation). ↩
23. Lupine Science, *Strategic Discovery Plan*, §6.3.2. ↩